

KYLE SHAFFER

2606A NW 57th Street, Seattle, WA 98107
414-517-7191 ◊ kyle.james.shaffer@gmail.com ◊ [Google Scholar](#)

EDUCATION

Master of Science, Information Science May 2015
University of North Carolina at Chapel Hill
School of Data & Information Sciences
Thesis: Predicting Speech Acts in MOOC Forum Posts Using Conditional Random Fields
Advisor: Dr. Jaime Arguello

Bachelor of Arts, Philosophy May 2010
University of Wisconsin - Eau Claire, *Magna Cum Laude*

RESEARCH INTERESTS

Society & Technology Studies, Misinformation, Social Media, Algorithmic Fairness, AI Ethics

PUBLICATIONS

Roemmele, M., **Shaffer, K.**, Olsen, K., Wang, Y. and DeNeefe, S. (2023) [AbLit: A Resource for Analyzing and Generating Abridged Versions of English Literature](#). *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics* (pp 3717-3733). Association for Computational Linguistics.

Shaffer, K., (2021) [Language Clustering for Multilingual Named Entity Recognition](#). *Findings of the Association for Computational Linguistics: EMNLP 2021* (pp 40-45). Association for Computational Linguistics.

Hodas, N., **Shaffer, K.**, Yankov, A., Corley, C., Anderson, A., Cheney, W. (2017) [Beyond Fine Tuning: Adding Capacity to Leverage Few Labels](#). *Proceedings of the 31st Conference on Neural Information Processing Systems* (pp 1-7). Association for Computing Machinery.

Volkova, S., **Shaffer, K.**, Jang, J., Hodas, N., (2017) [Separating Facts from Fiction: Linguistic Models to Classify Suspicious and Trusted News Posts on Twitter](#). *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics* (pp 647-653). Association for Computational Linguistics.

Phillips, L., **Shaffer, K.**, Arendt, D., Hodas, N., Volkova, S. (2017) [Intrinsic and Extrinsic Evaluation of Spatiotemporal Text Representations in Twitter Streams](#). *Proceedings of the 2nd Workshop on Representation Learning for NLP*. (pp 201-210). Association for Computational Linguistics.

Arguello, J. and **Shaffer, K.** (2015) [Predicting Speech Acts in MOOC Forum Posts](#). *Proceedings of the International AAAI Conference on Web and Social Media*, 9(1) (pp 2-11). Association for the Advancement of Artificial Intelligence.

Shaffer, K. (2010) [The Skeptic's Language Game: Does Sextus Empiricus Violate Normal Language-Use?](#) *Stance: An International Undergraduate Philosophy Journal*, 3(1) (pp 47-54).

INDUSTRY RESEARCH EXPERIENCE

Research Scientist, Natural Language Processing Sep 2020 – May 2026
Language Weaver (RWS Group)

- Conducted applied research in natural language processing and machine learning in areas such as document classification, named entity recognition, and text generation.
- Published findings in top natural language processing conference proceedings.

- Analyzed, cleaned, and standardized large-scale datasets for use in machine learning projects across the broader applied research team.

Senior Data Scientist

Dec 2018 – Sep 2020

Qntfy

- Trained and evaluated natural language processing models for investigating social media phenomena for research projects funded by DARPA and the Department of Defense.
- Implemented advanced deep learning techniques from research literature including transfer learning, multi-task learning, and multi-modal modeling.

Data & Applied Scientist II

Jun 2016 – May 2017

Pacific Northwest National Laboratory

- Developed and validated deep learning models for application areas in language and image analysis for federally funded research projects relevant to state and national security.
- Cleaned, standardized, and analyzed large-scale social media datasets for use in research at the intersection of machine learning and computational social science.
- Performed in-depth literature reviews and research of latest state-of-the-art methods and contributed to writing research papers for conference publications.

ACADEMIC RESEARCH EXPERIENCE

Research Assistant

August 2014 - May 2014

University of North Carolina at Chapel Hill

School of Data & Information Sciences

- Research assistant to Dr. Jaime Arguello developing machine learning models for automatically predicting student information needs in online course discussion forums.
- Cleaned, analyzed and formatted data for machine learning experiments, trained and evaluated models, and contributed literature review and methods section of paper.
- Findings presented in paper "Predicting Speech Acts in MOOC Forum Posts" accepted to International Conference on Web and Social Media 2015.

Research Assistant

August 2014 - May 2014

University of North Carolina at Chapel Hill

School of Medicine

- Research assistant to Dr. Carlton Moore to investigating natural language processing techniques for medical record abstraction.
- Findings presented at poster session in American Medical Informatics Association 2015 Symposium.

Data Science Fellow

June 2014 - August 2014

Georgia Institute of Technology

Data Science for Social Good Fellowship

- Fully funded fellowship for data science students for applied research projects focused on positive community outcomes for the Atlanta metro area.
- Built interactive data visualization dashboard for insights into crowd-sourced data on cycling conditions in Atlanta metro area collected through [Cycle Atlanta](#) mobile app.
- Designed web-app using combinations of frontend technologies including HTML, CSS, JavaScript, and D3.js visualization library.
- Cleaned, ingested, and analyzed data collected through smartphones for monitoring rides throughout the city of Atlanta.

Undergraduate Research Assistant

January 2014 - May 2014

University of Wisconsin-Eau Claire

Department of Linguistics

- Research assistant to Dr. Lynsey Wolter investigating Zeno Vendler's semantics and philosophy of language.
- Conducted extensive literature review and contributed paper investigating the intersection between Vendler's semantics and analytic philosophy of language.
- Findings presented at poster session in "Zeno Vendler and Semantics of Believe and Know."

TEACHING EXPERIENCE

Graduate Level Philosophy Tutor

Marquette University

November 2010 - December 2010

Tutoring Center

- Tutored graduate course in Epistemology for non-Philosophy grad students.
- Supplemented course lectures, assisted in understanding and interpretation of texts and philosophical concepts.
- Helped develop paper topics and provided feedback on drafts.

Philosophy Tutor

Marquette University

August 2010 - October 2010

Tutoring Center

- Tutored undergraduate course in Logic & Critical Reasoning.
- Provided feedback on paper proposals and drafts.
- Supplemented lectures with discussion sessions.

Undergraduate Teaching Assistant

University of Wisconsin-Eau Claire

January 2010 - May 2010

Department of Psychology

- Served as undergraduate teaching assistant for course in Cognitive Psychology for Dr. Allen Keniston.
- Graded student papers and provided feedback on paper proposals.

GRANTS, HONORS, AND AWARDS

Data Science For Social Good Fellowship

Georgia Institute of Technology

June 2014

- Summer fellowship for data science students for applied research projects focused on positive community outcomes for the Atlanta metro area.

Student Research Travel Grant

University of Wisconsin-Eau Claire

April 2010

Department of Philosophy & Religious Studies

- Student travel grant for presentation of paper at 5th Annual Undergraduate Conference in Philosophy, SUNY-Oneonta, April 15-17, 2010.

Student-Faculty Research Grant

University of Wisconsin-Eau Claire

January 2010

Department of Linguistics

- Research grant for student-faculty project and research paper "Zeno Vendler and the Semantics of *Know* and *Believe*".